

# Scope of the Formal Period-Removal Step Relative to PGVWC07 Periodic Decomposition

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## 1 Source theorem

Pérez-García, Verstraete, Wolf, and Cirac prove the periodic decomposition theorem for translation-invariant MPS with periodic boundary conditions [PGVWC07, Theorem 5, lines 849–858]. The theorem starts with a TI state  $\psi \in (\mathbb{C}^d)^{\otimes N}$  whose canonical TI MPS representation has a single block with  $D \times D$  matrices  $A_i$ . Let

$$\mathcal{E}(X) = \sum_i A_i X A_i^\dagger.$$

If  $\mathcal{E}$  has  $p$  eigenvalues of modulus 1, then:

1. if  $p$  divides  $N$ , the state  $\psi$  is a superposition of  $p$   $p$ -periodic states, each with an MPS representation of bond dimension  $D$ ;
2. if  $p$  does not divide  $N$ , then  $\psi = 0$ .

The proof uses the peripheral spectral structure of the CP map from Fannes–Nachtergaele–Werner: there are orthogonal projectors  $P_k$  cyclically permuted by the matrices  $A_i$ , and inserting  $\sum_k P_k$  into the trace splits  $\psi$  into the claimed  $p$ -periodic summands [PGVWC07, lines 860–880].

## 2 Formal boundary

Chapter 8 formalizes only the period-removal-by-blocking consequence needed for the canonical-form reduction: after blocking by a common multiple of the periods, the relevant cyclic sectors have primitive transfer maps.

This formal theorem is not the explicit PGVWC07 state-vector decomposition. It does not produce  $p$  separate  $p$ -periodic MPS summands for the original unblocked ring, nor does it state the dichotomy between  $p \mid N$  and  $\psi = 0$ . It supplies the primitive blocked sectors used by the later canonical-form development.

### 3 Required formalization

A source-faithful formalization of PGVWC07 Theorem 5 should state the one-block TI canonical hypothesis, the peripheral spectrum condition for  $\mathcal{E}$ , and the two conclusions above. The construction should exhibit the  $p$ -periodic summands obtained from the cyclic projectors  $P_k$ , including their bond dimension  $D$ , and should prove the vanishing statement when  $p \nmid N$ .

Until that theorem exists, Chapter 8 should cite the blocking theorem only as a period-removal input for primitive sectors, not as a formalization of the explicit periodic decomposition of the state vector.

### 4 Tracked consequence

The scope remark at the end of Chapter 8 records this boundary. It is a documented scope restriction, not a source-error claim. The open issue is issue #2278.

### References

- [PGVWC07] David Pérez-García, Frank Verstraete, Michael M. Wolf, and J. Ignacio Cirac. Matrix product state representations. *Quantum Information and Computation*, 7(5–6):401–430, 2007.